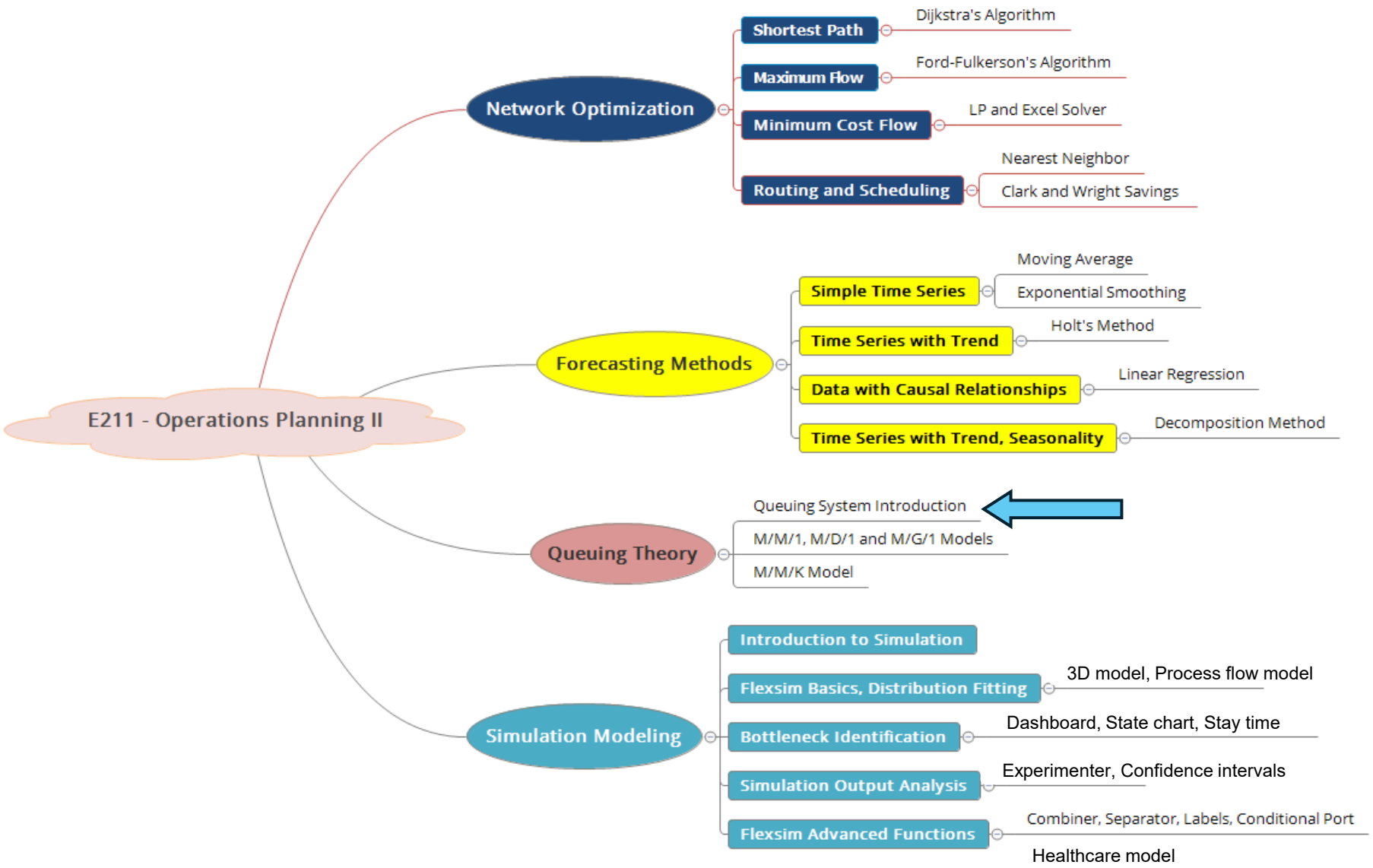
REPUBLIC POLYTECHNIC

Lesson 1
Introduction to Queuing Theory

E211 – Operations Planning II

REPUBLIC POLYTECHNIC

E211 Topic Tree



What is a Queuing System

REPUBLIC POLYTECHNIC

“Queuing is a way of life”

Long queues at Chagee opening

Queuing for CNY new notes

Queuing Theory

- It is estimated that the world spent a *billion* hours a year waiting in lines.

- Places we wait in line
  - Stores
  - Banks
  - Airports
  - Hotels
  - Traffic lights
  - Theme parks
  - Bus/ Train stations
  - Eateries

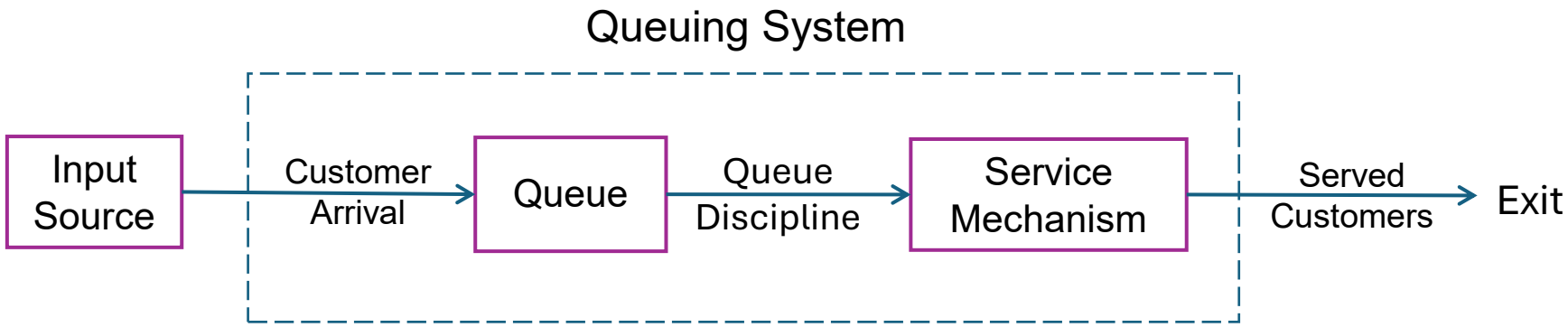
- When businesses plan their resources to manage their operations, one of the aims is to reduce customer waiting time and increase the number of customers served in an efficient manner.

- **Queuing Theory** is the <u>mathematical study of waiting lines or queues</u>.

Queuing System

- Any queues in the real world can be studied as a queuing system. A **Queuing System** is characterised by **three** basic components:
  - **Arrival**
    - Customers arrive according to some arrival pattern.
  - **Waiting Line or Queue**
    - Arriving customers may have to wait in one or more queues for service.
  - **Service**
    - Server(s)/ Counters providing service to customers

Basic Queuing Process



- **Customer arrivals** are generated over time by an input source.
- The customers enter the queuing system and join a **queue**.
- At certain times, a customer in the queue is selected for service by some rule known as the queue discipline.
- The required service is then performed by the **service mechanism**, after which the customer leaves the queuing system.

Terminologies

- **Arrival**
  - An entity (people or object) that arrives and requires service
- **Queue**
  - Waiting line
- **Queue discipline**
  - Rules for determining the order that arrivals receive service. E.g., FIFO, random order, priority
- **Phase**
  - Number of steps/ stages in the service; after getting service from one queue, move to the next queue for another service.
- **Channel**
  - Number of **parallel servers**

Why Queues Form (I)

- Waiting line forms because of **imbalance between demand for service and the capacity of the system** to provide the service.

- Customers usually arrive at unpredictable intervals (variability in arrival rate). When the **rate of arrival is higher than the rate of service** within a certain period, the capacity of the system gets overwhelmed, and queues start to form.

- The rate of producing service may also vary according to customer needs (variability in service rate) such that the rate of service is lower than the rate of arrival at times.

Why Queues Form (II)

- That is why in a queuing system, you should never match the average speed at which people arrive with the average speed at which they are served
- This will create long queues if **variability in arrival rate and service rate** are present.
  - Intervals between arrivals of people may be random
  - Service times may also be random
  - Randomness ensures that sometimes there will be no one waiting to be served; and at other times, a whole bunch of people will queue up.

Activity: Simulate and Observe

Run the queue simulation model.

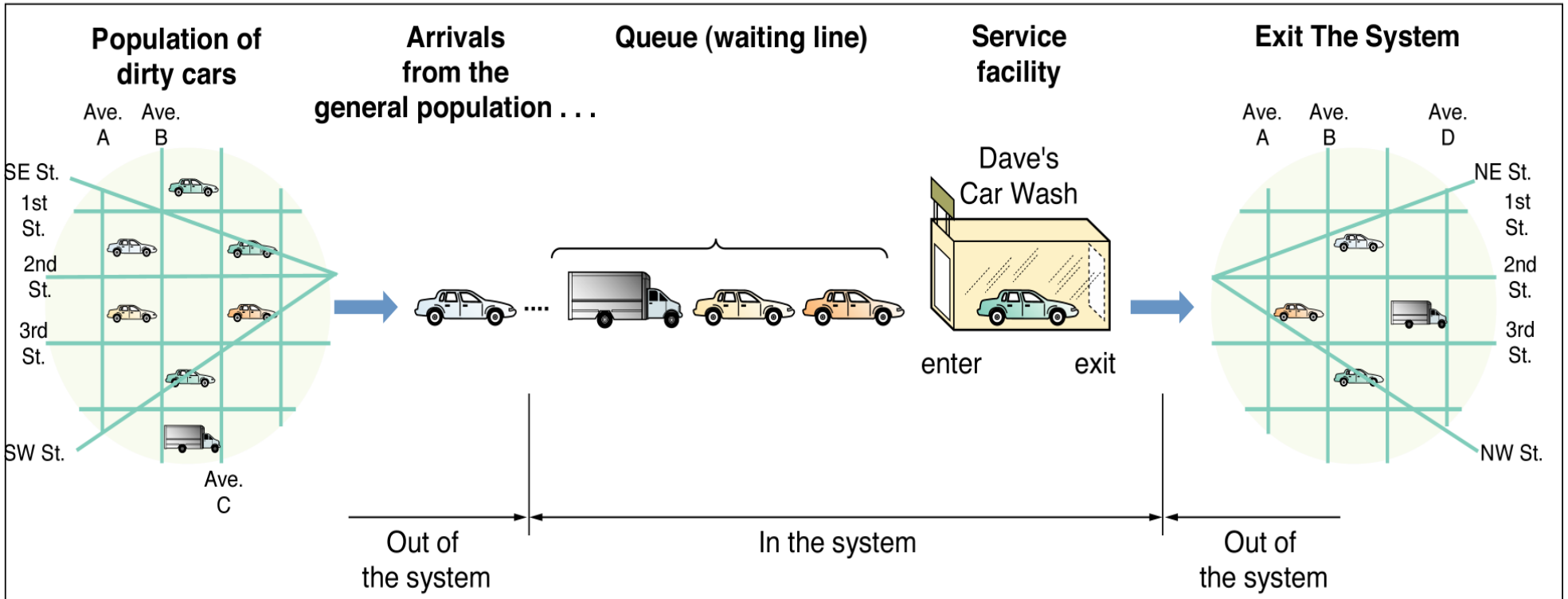
Queue Simulation.fsm

- Why do queues form in the simulation?
- What happens when the average arrival rate is greater than the average service rate?

Queuing System Characteristics

Example: Car-Wash Queuing System

Components of a Queuing System



Arrival Characteristics

- ❑ Size of arrival population
- ❑ Behavior of arrivals
- ❑ Statistical distribution of arrivals

Waiting-line Characteristics

- ❑ Limited vs. Unlimited
- ❑ Queue Discipline

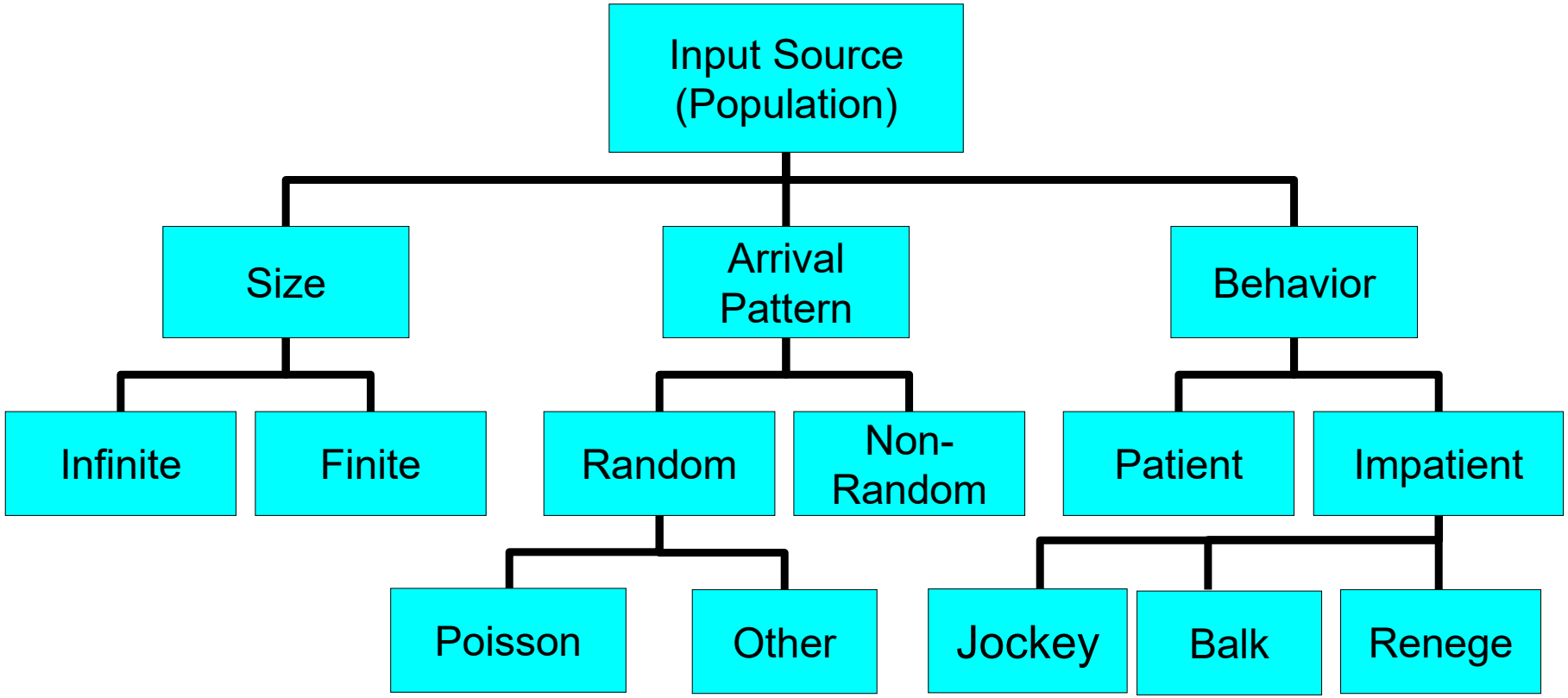
Service Characteristics

- ❑ Service design
- ❑ Statistical distribution of service

Arrival Characteristics

1. Size of Population
2. Arrival Pattern
3. Behavior of Arrivals

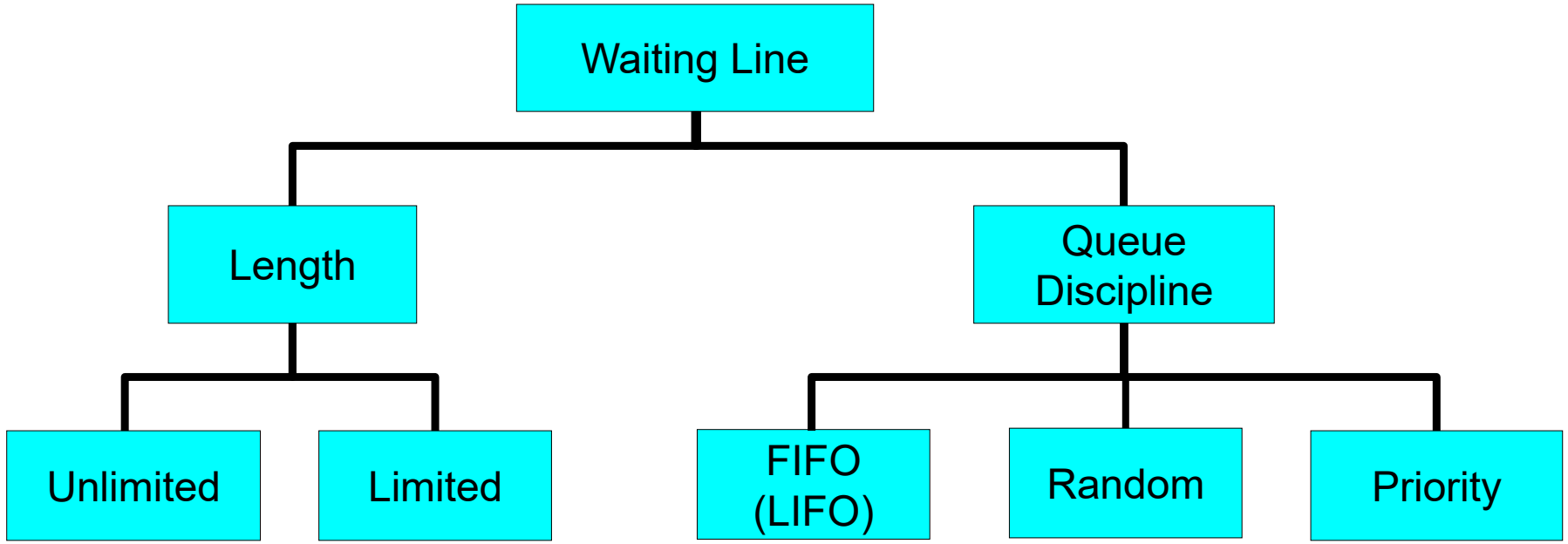
Balk – customers **refuse** to join the waiting line because it is too long for them
Renege – customers who are already in the waiting line but **leave** without being served
Jockey – customers **switch** between waiting lines



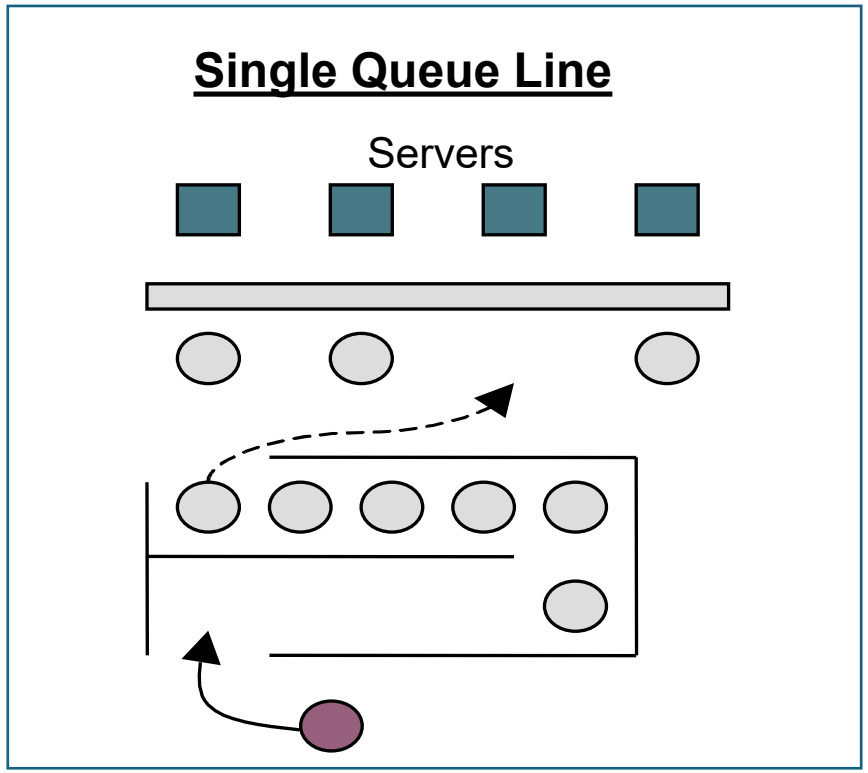
Queue Characteristics

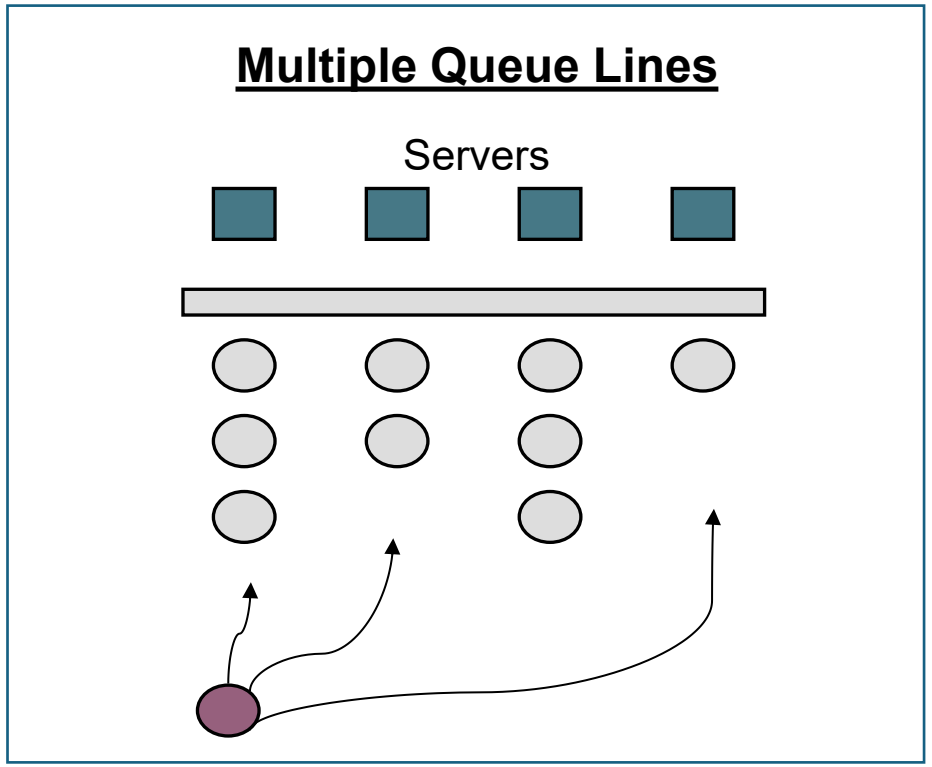
1. Length of Queue – physical restrictions
2. Queue Configuration – single or multiple lines
3. Queue Discipline – rule by which customers in the line are to receive service

E.g., FIFO (FCFS) – First-In-First-Out (First-Come-First-Serve)
LIFO – Last-In-First-Out



Single Queue Vs Multiple Queue Lines



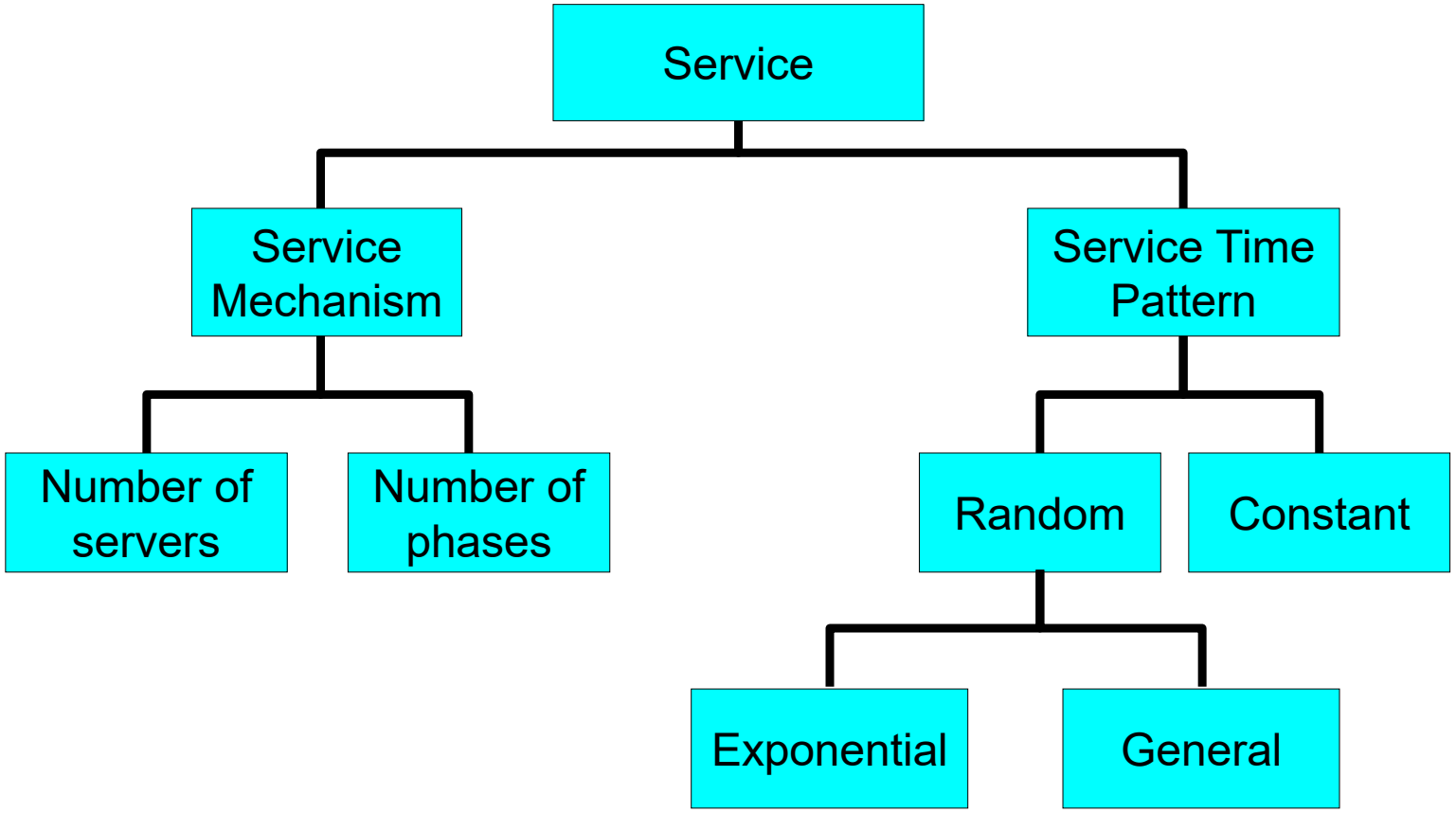


- **Guarantees fairness**
  FIFO applied to all arrivals
- **No customer anxiety regarding choice of queue**
- **Most efficient set up for minimizing time in queue**
- **Jockeying (line switching) is avoided**

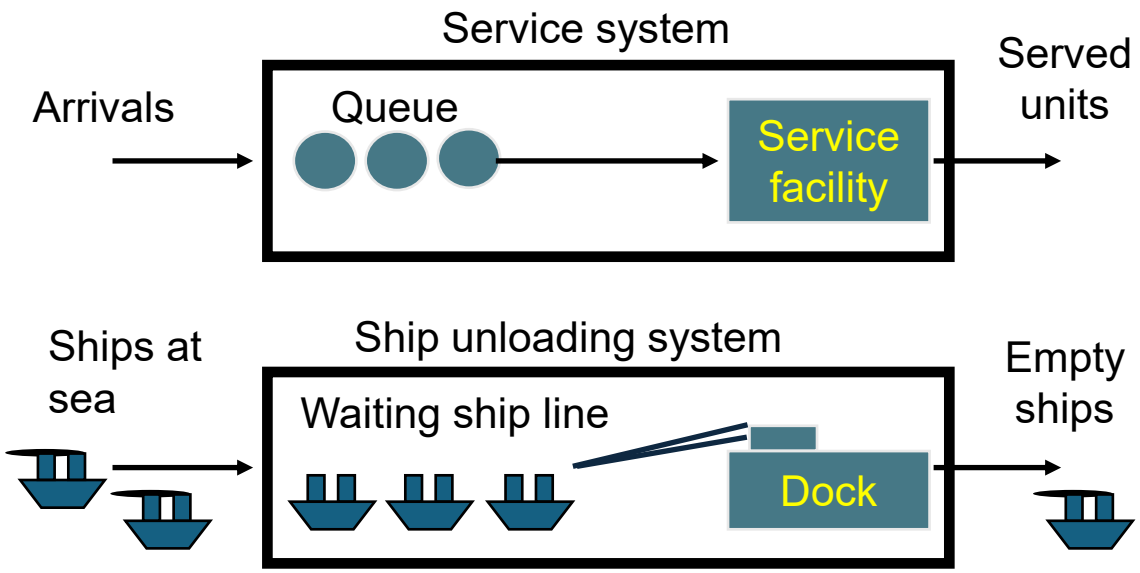
- **The service provided can be differentiated**
  Example: Supermarket express lanes
- **Labor specialization possible**
- **Customer has more flexibility**
- **Balking behavior may be deterred**
  Several medium-length lines are less intimidating than one very long line

Service Characteristics

1. Service mechanism
2. Service time pattern



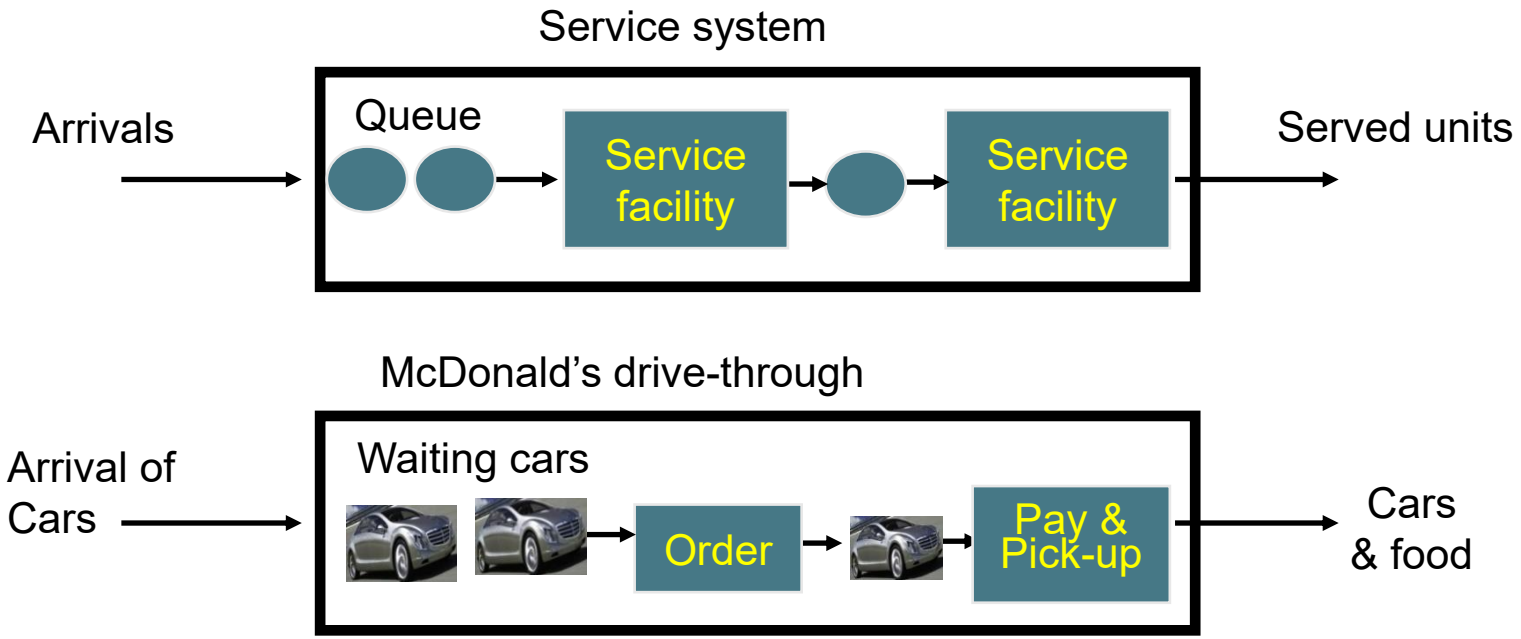
Service Mechanism – Single Server, Single Phase



Examples:

- Customers waiting in line to pay at a small 7-11 with single cashier.
- Ships waiting to unload at a single dock.

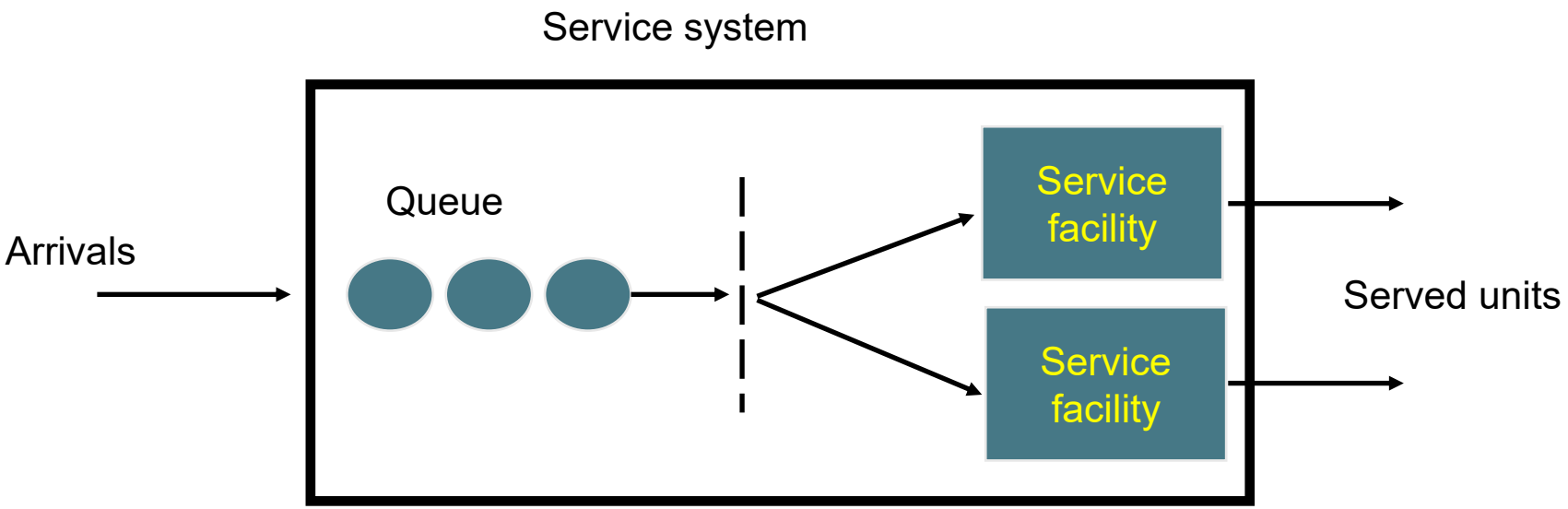
Service Mechanism – Single Server, Multi-Phase



Examples:

- Fast food drive-through with order and pick-up counters.
- Novelty photo taking booth (single booth) at events with 1 counter for photo taking and 1 counter for collection of print.

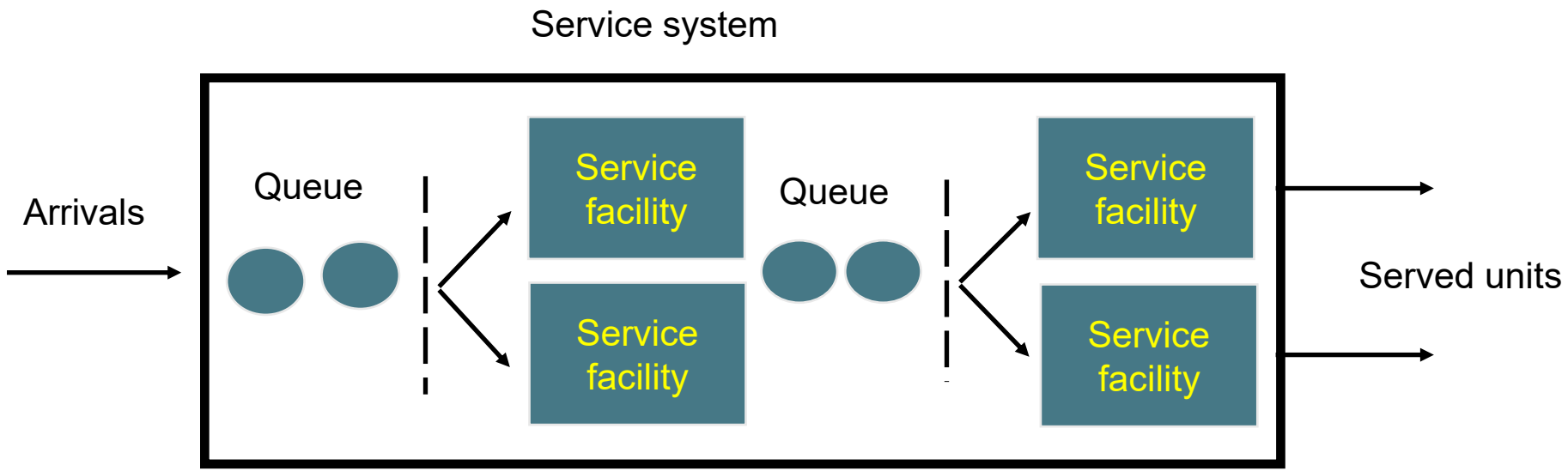
Service Mechanism – Multi-Server, Single Phase



Examples:

- Bank customers wait in line for one of several tellers.
- Mall customer wait in line for one of the several customer service staff.

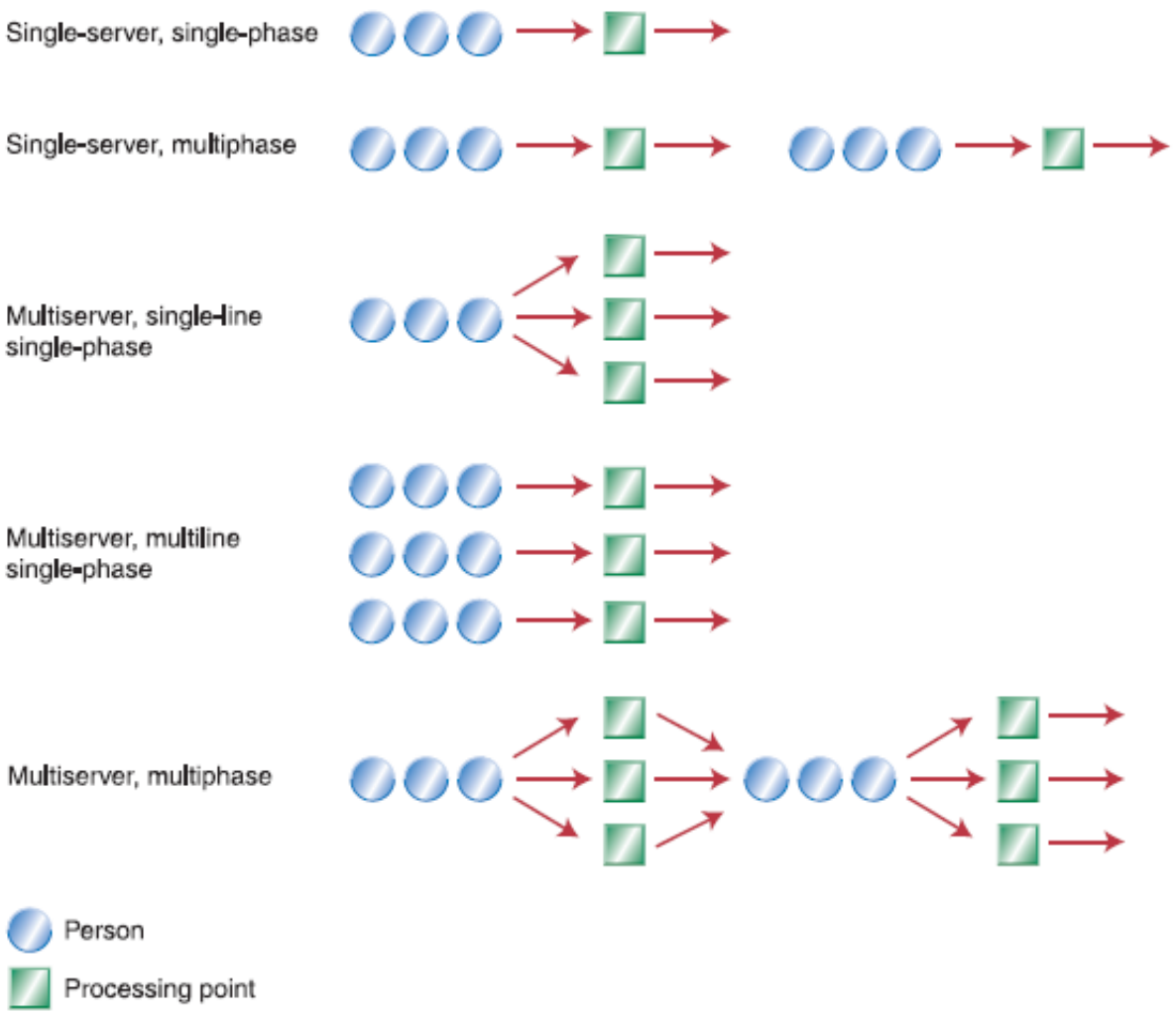
Service Mechanism – Multi-Server, Multi-Phase



Examples:

- At a laundromat, customers use one of several washers, then one of several dryers.
- Theme park - Queue to buy entrance tickets at the ticket booth, then queue to enter the theme park.

Summary of Service Mechanism



Activity: Describe the Queuing System Characteristics

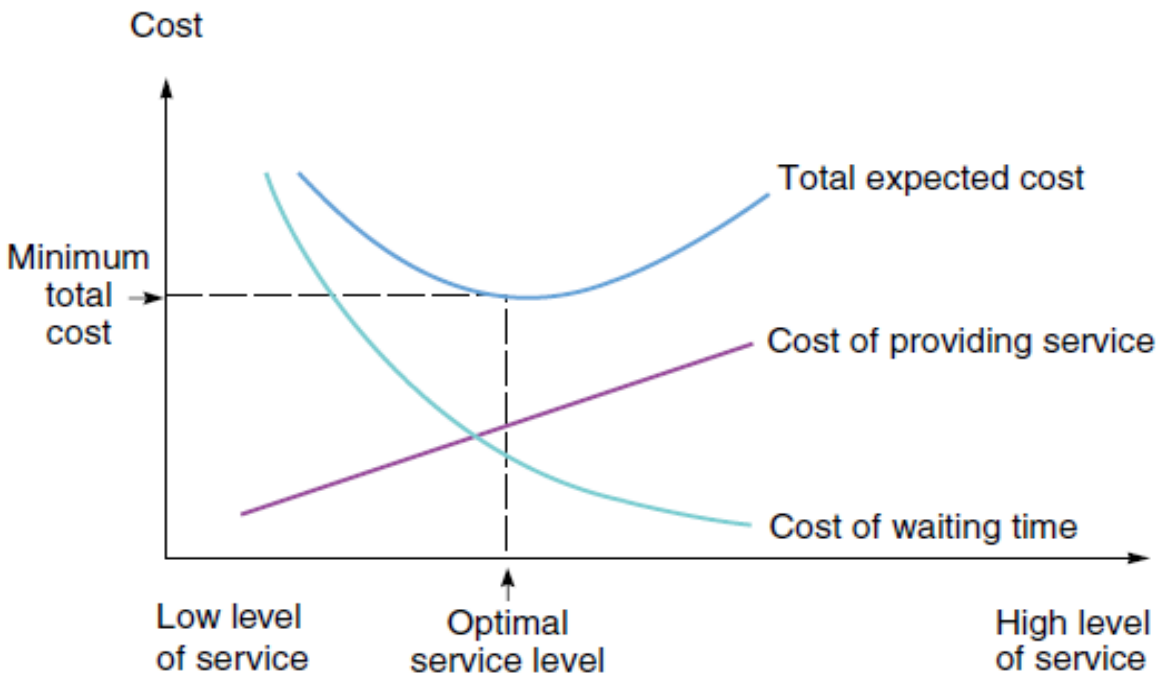
- RP IT Helpdesk
- Polyclinic
- Supermarket checkout
- Immigration clearance
- Your favourite drink stall/ shop

Characteristics to describe a queuing system

Arrival	Queue	Service
Population size	Queue capacity	Service time pattern
Arrival pattern	Queue configuration	Number of servers
Behaviour	Queue discipline	Number of phases

Queue Management Systems

Expected Cost of Queuing System



- **Cost of a queuing system (total expected cost)** consists of cost of providing service and cost of waiting.
- Initially, the cost of waiting in line is at a maximum when the level of service is at minimum.
- As level of service increases (e.g. more service counters), there is a reduction in the number of customers in the line and in their waiting times, which decreases waiting cost. However, this increases the cost of providing the service.
- The **optimal service level** corresponds to the point with the **minimum total cost**.

Queue Management Systems (I)

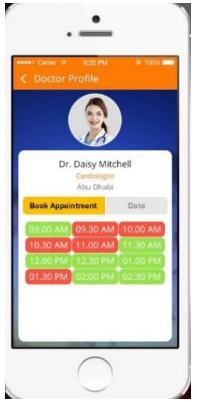
Queue management systems are deployed to manage queuing, control flow and improve customer experience by reducing the cost of waiting.

- Physical Barrier
  - Use of belt barriers or stanchions to enforce waiting lines

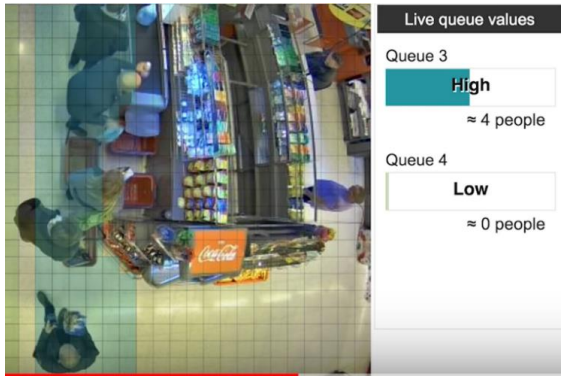
- Kiosk + Screen
  - Self service registration at clinics and food ordering at eateries
  - Payment and queue ticket issuance
  - Screens at waiting location display current queue number

Queue Management Systems (II)

- App/ Website
  - Enables appointment or reservation by customer
  - Virtual queue & scheduling - customers do not have to wait at location
  - Real-time queue info (estimated waiting time, position in queue)



- Smart monitoring systems
  - Mobile signal detection, AI video analytics to track crowd/ queue size & staff productivity
  - Enables real-time crowd control (busy large stores, security, etc.)



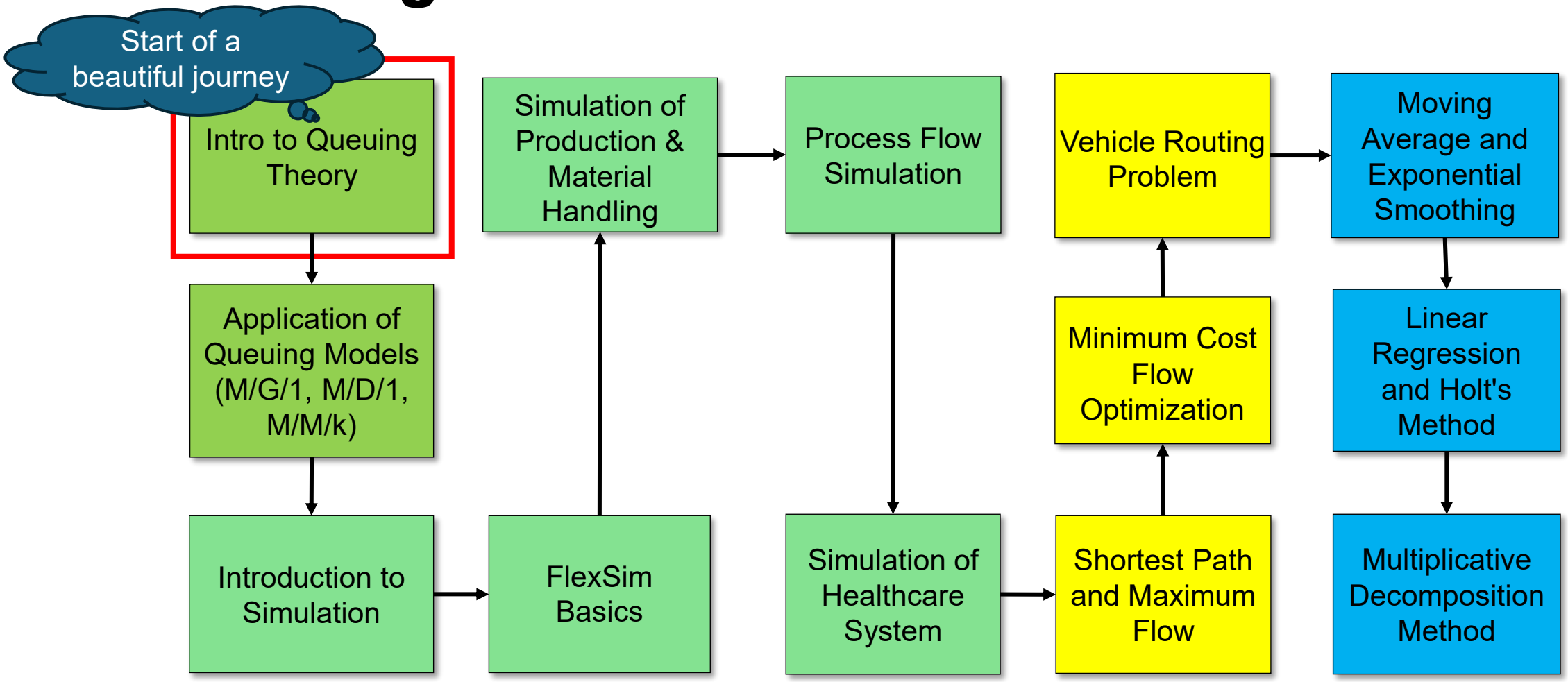
Queue Management Systems (III)

Benefits of queue management systems:

- Eliminate long physical waiting lines
- Boost staff productivity through customer self service
- Reduce customer dissatisfaction & increase business turnover
- Provide valuable customer data
- Improve business image

Learning Outcomes

At the end of the lesson, students should be able to:

- Explain why a queue forms in a process with limited resources.
- Describe the characteristics of a queuing system in terms of arrival, queue and service components.
- State the different types of queue management systems and their benefits.

E211 Progression